The position of the Centroid, Orthocentre, Circumcentre and Incentre differs depending on the type of triangles given. The following points will help us in locating and remembering these.

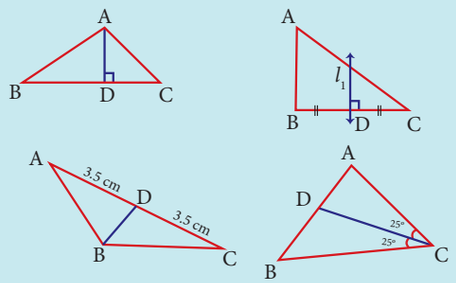
(i) For all types of triangles, Centroid (G) and Incentre (I) will be inside the triangle.

(ii) The Orthocentre (H) will be inside in an acute angled triangle, outside in an obtuse angled triangle and on the vertex containing 90° in a right angled triangle.

(iii) The Circumcentre (S) will be inside in an acute angled triangle, outside in an obtuse angled triangle and on the hypotenuse in a right-angled triangle.

Try these

Identify the type of segment required in each triangle:
(median, altitude, perpendicular bisector, angle bisector)



(i) AD =_____ (ii) l_1 =_____

(iii) BD =_____ (iv) CD =_____

Activity

1. By paper folding, find the centroid, orthocentre, circumcentre and incentre of an equilateral triangle. Do they coinside?
2. By paper folding, find the centroid (G), orthocentre(H), circumcentre (S) and incentre (I) of a triangle. Join G,H,S and I. Are they collinear?

Exercise 5.2

1. **Fill in the blanks:**

(i) If in a ΔPQR, $PR^2 = PQ^2 + QR^2$, then the right angle of ΔPQR is at the vertex _____.

(ii) If 'l' and 'm' are the legs and 'n' is the hypotenuse of a right angled triangle then, l^2 = _____.

(iii) If the sides of a triangle are in the ratio 5:12:13 then, it is _____.

(iv) The medians of a triangle cross each other at _____.

(v) The centroid of a triangle divides each medians in the ratio _____.

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